

Anish Kumar

 Github: github.com/Anish-ai
 LinkedIn: linkedin.com/in/anish-kumar-71779326a
 Portfolio Website: anish-k.me

 Email: aniskum59431@gmail.com
 Mobile: +91-9341027738

EDUCATION

Indian Institute of Technology (IIT), Patna CGPA: 7.34 **Aug 2023 – May 2027 (expected)**
Bachelor of Technology in Mathematics and Computing Patna, India

- Relevant Coursework:** Data Structures and Algorithms, Number Theory, Machine Learning, Object Oriented Programming, Database Management Systems, Probability and Statistics, Artificial Intelligence, Convex Optimizations, Computer Architecture

Senior Secondary (CBSE) Percentage: 91.2% **2023**

SKILLS

Core Concepts: Data Structures & Algorithms, Graph & Number Theory, Probability & Statistics, Operating Systems, DBMS, OOPs
Databases & Tools: SQL (PostgreSQL, MySQL), pgvector, Git, GitHub Actions, CI/CD, Docker, Kubernetes, Vercel, Firebase, MongoDB
Systems: Linux, Azure, Shell Scripting, Performance Optimization, Multithreading (basic)
Languages: C++, Python, Java, JavaScript, TypeScript, Go
Frameworks: React, React Native, Next.js, Node.js, FastAPI, Tailwind
Libraries: NumPy, Pandas, OpenCV, XGBoost, Random Forest

EXPERIENCE

Optum Global Solutions (UHG) Bengaluru, India **June 2026 – Present**
Software Engineering Intern

- Building the **Production Update Database Tool** using **Java, Python, and TypeScript** on **Azure** and **Kubernetes**, scaling to support over **1,000 internal teams** with automated SQL, PL/SQL, Oracle, and T-SQL query execution.
- Optimized system performance to efficiently parse **5,000+ complex queries in under 0.3 seconds**, significantly accelerating internal database workflows.
- Engineered an **offline data ingestion pipeline** incorporating advanced **drift and duplicate detection** mechanisms, utilizing **pgvector** for embedding storage and **MySQL** for relational data.
- Architected an agentic **Retrieval-Augmented Generation (RAG)** and LLM pipeline mapping user queries through a SQL parser, feature extractor, and rules engine to surface actionable plain-language summaries and secure rollback scripts.

TLE Eliminators Remote **May 2026 – Present**

Post-Contest Discussion (PCD) Head

- Led post-contest discussions for a large competitive programming community, recording and publishing **15+ live video breakdowns** of complex algorithmic problem solutions.
- Demonstrated strong technical communication and organizational skills by translating advanced problem-solving techniques and data structures into highly accessible educational content for diverse audiences.

ACHIEVEMENTS

- **Candidate Master (1963)** on **Codeforces** - **Profile Link**
- **Ranked 186th globally** in **Meta Hacker Cup 2025**, progressed to Round 3 - **Certificate**
- **4 Star (1839 rated)** on **CodeChef** - **Profile Link**
- **Global Rank 40** in **CodeChef Starters 219**
- **Achieved rank 337** among **38,278** participants in **Codeforces Round 1027 (Div. 3)**
- **Reached Round 2** in **The Big Code Challenge by Google 2026**
- **Selected for Summer of Bitcoin 2025**; contributed to **ZeusLN Wallet** in Linux environment by implementing and validating on-chain message signing and verification using **yarn test**, **lint**, **tsc**, and **prettier**
- **Ranked top 0.76%** in **JEE Advanced** among 1.1M+ candidates

PROJECTS

Volatility Curve Prediction (Financial Time-Series) | NK Securities Research Hackathon **Jun 2025**

- Secured **6th rank** among 6,000+ submissions in a highly competitive quantitative finance and algorithmic problem-solving hackathon (**Certificate**).
- Developed and optimized machine learning models to analyze financial time-series data, accurately predicting volatility curves and showcasing the ability to derive actionable insights from complex datasets.

Facility Routing Engine |  Github |  Live Website **Mar 2026**

- Tech Stack:** Next.js, React 19, TypeScript, Zustand, Canvas API, A*/Dijkstra, Vercel
- Modeled indoor navigation as a **weighted graph**, implementing **A* and Dijkstra algorithms** for optimal pathfinding.
- Optimized pathfinding computations by reducing redundant graph traversals, improving query efficiency under dynamic constraints.
- Implemented **emergency-aware pathfinding** for fire, medical, security, smoke, and hazmat scenarios with dynamic rerouting around hazardous zones.
- Added constraint-based routing rules such as fire-safe routing, emergency-zone avoidance, and stair preference over lifts during fire scenarios.

OneIITP (App Development) |  PlayStore |  AppStore |  OneAttend **Aug 2024 – Present**

- Built and launched IIT Patna's official student app for **2,000+ active users**, increasing engagement by **40%** and generating **15,000+ reads**, **1,500+ downloads**, and a **4.9/5** rating.
- Developed a **QR-based anti-proxy attendance system** with auto-refreshing codes, cutting proxy attendance by **90%**, and created the **OneAttend** portal for professor-side attendance tracking and analytics.
- Integrated **Azure authentication** and **Google Sheets API** to sync real-time bus schedules and campus data, improving convenience for students and staff by **30%**.

POSITIONS OF RESPONSIBILITY & VOLUNTEER EXPERIENCE

Dev Lead for Cyber Security Problem Statement: <i>ICTC 3.0</i>	Mar 2025 – Apr 2025
–  <i>Deployed Website Solution</i>  <i>Problem Statement</i>	
Overall Tech Coordinator <i>Anwesha, IIT Patna</i>	Apr 2025
Dev & Design Team <i>Inter IIT Cult 2024, IIT Patna</i>	Dec 2024

CERTIFICATIONS

Certificate of Achievement: Murf AI Coding Challenge 2025 Certificate Link	Jun 2025
Google Cloud Computing Foundations: Infrastructure in Google Cloud Certificate Link	Oct 2023